Annotated Bibliography

Raelyn Brooks

November 11, 2026

References

[1] Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai–tibet plateau and high mountain regions. 2025. Unpublished thesis.

This thesis investigates environmental contaminants in high-altitude water and snow samples using high-resolution mass spectrometry. It highlights the importance of computational methods for molecule identification and demonstrates how suspect screening and advanced data analysis techniques can reveal pollutants in remote ecosystems.

[2] Mónika Adányi, Manuel A. Andino-Enriquez, Masho Hilawie Belay, Alessia Fabbris, Arianna Ghignone, Paola Calza, Emilio Marengo, and Elisa Robotti. Widespread emerging and legacy contaminants across alpine snow and surface waters revealed by non-target analysis. *Environmental Pollution*, 407:128754, 2026.

This study utilizes LC-HRMS and non-target screening to investigate emerging and legacy organic contaminants in alpine snow and surface waters from the Monte Bianco region. It demonstrates the analytical challenge of parsing complex environmental chemical fingerprints comprising hundreds of uncharacterized features. In this project, this paper provides the real-world environmental application context—demonstrating the exact type of high-altitude water and snow sample datasets that require localized, transparent molecular networking and feature prioritization workflows.

[3] Francesco Asnicar, Andrew Maltez Thomas, Andrea Passerini, Levi Waldron, and Nicola Segata. Machine learning for microbiologists. *Nature Reviews Microbiology*, 22(4):191–205, 4 2024.

This article reviews the application of machine learning techniques in microbiology, highlighting key advancements and future directions. It emphasizes the potential of machine learning to enhance our understanding of microbial communities, their functions, and their interactions with the environment. The insights gained from this review will inform my research on the use of

machine learning for analyzing complex microbiome data, particularly in the context of environmental samples.

[4] Albert-László Barabási and Márton Pósfai. *Network Science*. Cambridge University Press, 2016.

This textbook establishes the fundamental mathematical and topological principles of network science, exploring graph structures, connectivity, and complex systems. In this project, it provides the theoretical framework for representing mass spectrometry feature data as graph networks, where molecular features act as nodes and cosine similarity scores function as weighted edges.

[5] Roberto J. Bayardo, Yiming Ma, and Ramakrishnan Srikant. Scaling up all pairs similarity search. In *Proceedings of the 16th International Conference on World Wide Web*, pages 131–140, 2007.

This paper presents high-performance algorithms for computing all-pairs similarity across large sets of sparse, high-dimensional vectors. It offers algorithmic strategies relevant to optimizing backend cosine similarity matrix calculations for mass spectral datasets.

[6] Wout Bittremieux, Robin Schmid, Florian Huber, Justin J. J. van der Hooft, Mingxun Wang, and Pieter C. Dorrestein. Comparison of cosine, modified cosine, and neutral loss based spectrum alignment for discovery of structurally related molecules. *Journal of the American Society for Mass Spectrometry*, 33(9):1733–1744, 2022.

This study compares different spectrum alignment methods for the discovery of structurally related molecules in mass spectrometry data.

[7] Jules M. Blais, Michael R. Rosen, and John P. Smol. Environmental contaminants: Using natural archives to track sources and long-term trends of pollution. 2015.

This book discusses the use of natural archives, such as sediments and ice cores, to track the sources and long-term trends of environmental pollution, providing valuable insights into the history of contaminants in the environment.

[8] Vincent Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008:P10008, 10 2008.

This paper introduces the Louvain community detection algorithm for efficiently identifying clustered substructures within large networks. In this project, it serves as the algorithmic basis for automatically grouping related molecular features and chemical families within the interactive network visualization.

[9] Béla Bollobás. *Random Graphs*, volume 73 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2nd edition, 2001.

[10] Ulrik Brandes and Thomas Erlebach. *Network Analysis: Methodological Foundations*, volume 3418 of *Lecture Notes in Computer Science*. Springer-Verlag, Berlin, Heidelberg, 2005.

[11] Jordi. Catalan. *Tracking Long-Range Atmospheric Transport of Trace Metals, Polycyclic Aromatic Hydrocarbons, and Organohalogen CompoundsUsing Lake Sediments of Mountain Regions*. Springer, Dordrecht, 2015.

This book chapter discusses the use of lake sediments from mountain regions as archives for tracking long-range atmospheric transport of various pollutants, including trace metals, polycyclic aromatic hydrocarbons, and organohalogen compounds. It emphasizes the importance of understanding atmospheric transport processes and their impact on remote ecosystems, particularly in the context of global environmental change. The chapter provides valuable insights for my research on the fate and transport of organic contaminants in high-altitude environments, highlighting the need for integrated approaches that combine field measurements, laboratory analyses, and modeling efforts.

[12] Alex Chao, Jeffrey M. Minucci, Troy M. Ferland, E. Tyler Carr, Greg Janesch, Safia Rizwan, Heather D. Whitehead, Tommy Cathey, Shirley Pu, Laura D. Brunelle, Angela L. Batt, Jon R. Sobus, and Antony J. Williams. Prioritizing chemical candidates from non-targeted analysis using metadata, spectral similarity, and hazard scoring within INTERPRET NTA. *Analytical Chemistry*, 97(29):15904–15912, 2025.

This work introduces the INTERPRET NTA computational framework, which integrates spectral similarity, metadata, and hazard scoring to prioritize chemical candidates in HRMS datasets.

[13] B. Coenen and P. Edminston. Detection identification of designer benzodiazepines via mass spectrometry-based molecular networking. 2026. Unpublished thesis.

This undergraduate thesis explores the use of mass spectrometry-based molecular networking for the detection and identification of designer benzodiazepines, highlighting the potential of this approach for forensic and analytical applications.

[14] Nicola Dalla Valle, Mar Garcia-Aloy, Peter Robatscher, Pietro Franceschi, and Michael Oberhuber. Improving spectral similarity and molecular network reliability through noise signal filtering in ms/ms spectra. *Analytical Chemistry*, 97(29):15873–15882, 7 2025. PMID: 40673560; PMCID: PMC12311886.

This study presents a method for improving spectral similarity and molecular network reliability by filtering noise signals in MS/MS spectra.

[15] Gillian L. Daly and Frank Wania. Organic contaminants in mountains. *Environmental Science & Technology*, 39(2):385–398, 2005.

This article reviews the occurrence and fate of organic contaminants in mountainous regions, highlighting the unique challenges posed by these environments for pollution monitoring and management.

[16] Priscillia Ann David, Anis Irfan Norazhar, and Mohamad Shazeli Che Zain. Integrating lc-ms/ms and molecular networking for advance analysis and comprehensive flavonoids annotation. *Journal of Separation Science*, 48(7):e70230, 2025.

This review examines the integration of LC-MS/MS and molecular networking (including classical and feature-based workflows) to streamline the structural annotation and dereplication of complex metabolomic datasets. It highlights how spectral similarity matching and network visualization help resolve structural isomers and cluster related molecular families. In this project, it provides methodological support for utilizing tandem mass spectrometry data structures and cosine-similarity-based graph architectures to map chemical fragmentation relationships.

[17] Deborah A. Davidson, Andrew C. Wilkinson, Jules M. Blais, Lynda E. Kimpe, Karen M. McDonald, and David W. Schindler. Orographic cold-trapping of persistent organic pollutants by vegetation in mountains of western canada. *Environmental Science & Technology*, 37(2):209–215, 2003.

This study highlights the role of orographic cold-trapping in the accumulation of persistent organic pollutants in mountainous regions, emphasizing the need for further research on the environmental impacts of these contaminants in alpine ecosystems.

[18] Reinhard Diestel. *Graph Theory*, volume 173 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1st edition, 1997.

A comprehensive and mathematically rigorous graduate-level introduction to modern graph theory. It balances foundational theorems with recent developments, covering topics such as matching, connectedness, planarity, colorings, flows, and extremal graph theory. Widely regarded as a standard reference text in the field.

[19] S. N. Dorogovtsev and J. F. F. Mendes. *Evolution of Networks: From Biological Nets to the Internet and WWW*. Oxford University Press, Oxford, 2003.

One of the earliest and most influential technical texts on the statistical physics of complex networks. It explores the foundational principles governing the organization, growth, and topology of real-world systems, ranging from biological nets to the Internet and the World Wide Web.

[20] Madeleine Ernst, Kyo Bin Kang, Andrés Mauricio Caraballo-Rodríguez, Louis-Felix Nothias, Joe Wandy, Christopher Chen, Mingxun Wang, et al. MolNetEnhancer: Enhanced molecular networks by integrating metabolome mining and annotation tools. *Metabolites*, 9(7):144, 2019.

MolNetEnhancer integrates molecular networking with multiple annotation tools to improve structural interpretation of complex metabolomics datasets.

[21] Christine M. Fisher, Katherine T. Peter, Seth R. Newton, Andrew J. Schaub, and Jon R. Sobus. Approaches for assessing performance of high-resolution mass spectrometry-based non-targeted analysis methods. 414(22):6455–6471, 2022.

Discusses performance assessment for non-targeted analysis using HRMS, including sample classification, chemical identification, and quantitation. Relevant to evaluating identification confidence, methodological reliability, and the performance of computational NTA workflows.

[22] Ari M. Frank, Nuno Bandeira, Zhouxin Shen, Stephen Tanner, Steven P. Briggs, Richard D. Smith, and Pavel A. Pevzner. Clustering millions of tandem mass spectra. *Journal of Proteome Research*, 7(1):113–122, 2008.

[23] Rebekah Gray, Eve Painter, Jameson W. Sprankle, Alex D. Crawford, Jillian J. Morrison, Marian Frazier, and Jennifer A. Faust. Suspect screening for pesticides in rain and snow using liquid chromatography high-resolution mass spectrometry. *Atmospheric Environment*, 291:119389, 2022.

This study employs high-resolution mass spectrometry (HRMS) for suspect screening of pesticides in environmental samples, demonstrating the method's effectiveness in identifying contaminants in complex matrices like rain and snow.

[24] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using networkx. In Gërard Varoquaux, Travis Vaught, and Jarrod Millman, editors, *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pages 11–15, 2008.

This paper introduces NetworkX, a Python software package designed for the creation, manipulation, and study of complex network structures, utilizing a dictionary-of-dictionaries data structure and integrating with NumPy and SciPy libraries. In this project, it provides the core computational graph data structures and algorithmic foundation for constructing, analyzing, and processing molecular networks from LC-HRMS feature data[cite: 1, 2, 3].

[25] B. M. J. Herbert, S. Villa, and C. J. Halsall. Chemical interactions with snow: Understanding the behavior and fate of semi-volatile organic compounds in snow. *Ecotoxicology and Environmental Safety*, 63(1):3–16, 2006.

This article discusses the behavior and fate of semi-volatile organic compounds in snow, emphasizing the importance of understanding these interactions for assessing environmental contamination.

[26] Johannes Hoffmann, Tobias Ternent, Juan Antonio Vizca´ino, Nils Hoffmann, Gerhard Mayer, Andreas Bertsch, Oliver Kohlbacher, Martin Eisenacher, Christoph Steinbeck,

Julian Uszkoreit, et al. mztab-M: A data standard for sharing quantitative results in mass spectrometry metabolomics. *Analytical Chemistry*, 91(8):5330–53310, 2019.

Introduces mzTab-M, a standardized, lightweight text-based data format specifically designed for reporting quantitative results, metadata, and small molecule/metabolite identifications from mass spectrometry-based metabolomics experiments. It facilitates interoperability and data deposition across public metabolomics repositories.

[27] Hisayuki Horai, Masanori Arita, and Takaaki Nishioka. Comparison of esi-ms spectra in massbank database. In *2008 International Conference on BioMedical Engineering and Informatics*, volume 2, pages 853–857, 2008.

This paper presents a comparison of ESI-MS spectra in the MassBank database, highlighting the importance of spectral libraries for metabolomics research.

[28] Henriette Hupatz, Ingrid Rahu, Wen-Chao Wang, et al. Critical review on in silico methods for structural annotation of chemicals detected with lc/hrms non-targeted screening. *Analytical and Bioanalytical Chemistry*, 417(2):473–493, 1 2025.

This article critically evaluates in silico methods for the structural annotation of chemicals identified through non-targeted screening techniques. It discusses the challenges and advancements in the field, providing insights into the potential of these methods for improving the identification and characterization of environmental contaminants. The review highlights the importance of integrating computational approaches with experimental data to enhance the reliability of chemical assessments in environmental studies.

[29] Mathieu Jacomy, Tommaso Venturini, Sebastien Heymann, and Mathieu Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the Gephi software. *PLOS ONE*, 9(6):e98679, 2014.

This paper details the ForceAtlas2 continuous force-directed layout algorithm, which positions nodes based on network connectivity to reveal structural clusters and hubs. It provides a methodological reference for the graph rendering and visualization mechanics used to organize chemical families and unknown transformation products in the software interface.

[30] Mikko Katajamaa, Jarkko Miettinen, and Matej Orešič. MZmine: toolbox for processing and visualization of mass spectrometry based molecular profile data. *Bioinformatics*, 22(5):634–636, 2006.

Introduces MZmine, an open-source Java-based software toolbox designed for the automated processing, alignment, and visualization of high-throughput mass spectrometry-based molecular profiles. It laid a foundational framework for data preprocessing in untargeted metabolomics and proteomics workflows.

[31] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and Oliver Fiehn. Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

This review explains how small molecules are identified using high-resolution MS/MS techniques and spectral libraries. It highlights computational approaches and shared databases that assist in analyzing complex environmental and biological samples.

[32] Aliz Kiss and Elke Fries. Occurrence of benzotriazoles in the rivers main, hengstbach, and hegbach (germany). *Environmental Science and Pollution Research*, 16(6):702–710, 2009.

This study investigates the occurrence of benzotriazoles in various rivers in Germany, highlighting the need for monitoring these contaminants in aquatic environments.

[33] Amy J. Ko, Robin Abraham, Laura Beckwith, Alan Blackwell, Margaret Burnett, Martin Erwig, Chris Scaffidi, Joseph Lawrance, Henry Lieberman, Brad Myers, Mary Beth Rosson, Gregg Rothermel, Mary Shaw, and Susan Wiedenbeck. The state of the art in end-user software engineering. *ACM Comput. Surv.*, 43(3), 4 2011.

This paper examines software engineering principles tailored to end-user developers and domain experts. It supports the project's philosophy of designing transparent, localized computational tools that empower domain scientists to execute complex data workflows without relying on black-box platforms.

[34] Jeremy Koelmel, Nicholas Kroeger, Emily Gill, Candice Ulmer Holland, John Bowden, Rainey Patterson, Richard Yost, and Timothy Garrett. Expanding lipidome coverage using lc-ms/ms data-dependent acquisition with automated exclusion list generation. *Journal of The American Society for Mass Spectrometry*, 28:1–10, 2017.

This paper presents a strategy for expanding mass spectrometry feature coverage using data-dependent acquisition combined with automated exclusion list generation to manage redundancy and enhance detection in complex analytical matrices. It underscores the value of automated data preprocessing and filtering pipelines, providing methodological context for streamlining complex LC-HRMS workflows and feature prioritization.

[35] Panos Louridas. Review of network science by albert-làaszlò barabàasi. *SIGACT News*, 49(2):10–13, 6 2018.

This review of Barabási's foundational work highlights the interdisciplinary evolution of network science, bridging computer science, mathematics, and domain-specific applications. It provides helpful context for the thesis literature review, emphasizing how graph-theoretic tools can be successfully adapted to non-traditional computational domains like analytical chemistry.

[36] Zhiyuan Ma, Shunlong Meng, Xin Gao, Hui Wan, Longxiang Fang, Lei Zhu, Zhonghua Li, Yuxiao Gao, Liping Qiu, Bo Rong, and Chao Song. Human activity driven spatial heterogeneity of pollutants in a highland river basin revealed by non-target screening and improved pbt evaluation: A case study from the yalong river on the qinghai-tibet plateau, china. *Journal of Hazardous Materials*, 496:139493, 2025.

This article provides valuable insights into the complex interactions between human activities and pollutant distribution in river basins. The use of advanced screening techniques and ecological assessments highlights the importance of tailored approaches for effective environmental management. The findings underscore the need for ongoing research in this area to address emerging contaminants and their impacts on aquatic ecosystems.

[37] Yu A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(4):824–836, 4 2020.

This paper introduces Hierarchical Navigable Small World (HNSW) graphs for efficient approximate nearest neighbor search in high-dimensional vector spaces. In this project, it provides the algorithmic foundation for optimizing high-dimensional spectral vector comparisons, enabling rapid cosine similarity calculations and scalable pairwise distance indexing across large environmental LC-HRMS datasets.

[38] K. E. Manz, A. Feerick, J. M. Braun, Y. L. Feng, A. Hall, J. Koelmel, C. Manzano, S. R. Newton, K. D. Pennell, B. J. Place, K. J. Godri Pollitt, C. Prasse, and J. A. Young. Non-targeted analysis (NTA) and suspect screening analysis (SSA): A review of examining the chemical exposome. *Journal of Exposure Science & Environmental Epidemiology*, 33(4):524–536, 2023.

This review examines the use of HRMS-based non-targeted analysis and suspect screening for identifying environmental contaminants in air, water, soil, and human samples. It highlights computational workflows and current limitations in exposome research.

[39] Alex Martelli. *Python in a Nutshell*. O'Reilly Media, Inc., Sebastopol, CA, 2003.

A concise desktop quick reference guide for Python programmers, covering core language syntax, object-oriented features, standard library modules, and practical utility functions across the language ecosystem.

[40] Torsten Meyer and Frank Wania. Organic contaminant amplification during snowmelt. *Water Research*, 42(8):1847–1865, 2008.

This article reviews the behavior of organic contaminants during snowmelt, emphasizing the need for further research to understand their release mechanisms and impacts on water quality.

[41] M. E. J. Newman. The structure and function of complex networks. *SIAM Review*, 45(2):167–256, 2003.

A seminal and highly influential review paper in network science. It provides a comprehensive synthesis of empirical network properties across sociology, biology, and technology (such as small-world topology, scale-free degree distributions, and clustering), while detailing the mathematical models, algorithms, and structural metrics used to analyze them.

[42] Stephanie Nino-Suastegui, Eve Painter, Jameson W. Sprankle, Jillian J. Morrison, Jennifer A. Faust, and Rebekah Gray. Non-targeted analysis and suspect screening of organic contaminants in temperate snowfall using liquid chromatography high-resolution mass spectrometry. *Environmental Research*, 266:120494, 2025.

This study uses LC-HRMS and in silico analysis to identify organic contaminants deposited in snowfall across the United States, demonstrating the power of computational tools for suspect screening and compound identification.

[43] Louis-Félix Nothias, Daniel Petras, Robin Schmid, Kai Dührkop, Johannes Rainer, Abinesh Sarvepalli, Ivan Protsyuk, Madeleine Ernst, Hiroshi Tsugawa, Markus Fleischauer, Fabian Aicheler, Alexander A. Aksenov, Oliver Alka, Pierre-Marie Allard, Aiko Barsch, Xavier Cachet, Andres Mauricio Caraballo-Rodriguez, Ricardo R. Da Silva, Tam Dang, Neha Garg, Julia M. Gauglitz, Alexey Gurevich, Giorgis Isaac, Alan K. Jarmusch, Zdeněk Kameník, Kyo Bin Kang, Nikolas Kessler, Irina Koester, Ansgar Korf, Audrey Le Gouellec, Marcus Ludwig, Martin H. Christian, Laura-Isobel McCall, Jonathan McSayles, Sven W. Meyer, Hosein Mohimani, Mustafa Morsy, Oriane Moyne, Steffen Neumann, Heiko Neuweger, Ngoc Hung Nguyen, Melissa Nothias-Esposito, Julien Paolini, Vanessa V. Phelan, Tomáš Pluskal, Robert A. Quinn, Simon Rogers, Bindesh Shrestha, Anupriya Tripathi, Justin J. J. van der Hooft, Fernando Vargas, Kelly C. Weldon, Michael Witting, Heejung Yang, Zheng Zhang, Florian Zubeil, Oliver Kohlbacher, Sebastian Böcker, Theodore Alexandrov, Nuno Bandeira, Mingxun Wang, and Pieter C. Dorrestein. Feature-based molecular networking in the GNPS analysis environment. *Nature Methods*, 17(9):905–908, 2020.

Introduces feature-based molecular networking (FBMN) within the Global Natural Products Social Molecular Networking (GNPS) framework. This method integrates untargeted LC-MS/MS data processing tools with molecular networking to resolve isomers, quantify structural variants, and account for ion mobility spectrometry data, significantly expanding the scope of mass spectrometry-based metabolomics and natural product discovery.

[44] Florent Olivon, Gwendal Grelier, Fanny Roussi, Marc Litaudon, and David Touboul. MZmine 2 data preprocessing to enhance molecular networking reliability. *Analytical Chemistry*, 89(15):7836–7840, 2017.

This study improves molecular networking workflows through data preprocessing using MZmine 2, demonstrating how careful data processing can improve clustering and compound annotation accuracy.

[45] Robert Pienta, Acar Tamersoy, Hanghang Tong, Alex Endert, and Duen Horng (Polo) Chau. Interactive querying over large network data: Scalability, visualization, and interaction design. In *Companion Proceedings of the 20th International Conference on Intelligent User Interfaces*, IUI '15 Companion, page 61–64, New York, NY, USA, 2015. Association for Computing Machinery.

This paper explores scalable graph visualization, data mining, and interactive design principles for navigating complex network structures. It directly supports the user interface and interaction design goals of the custom software, informing how chemists can interactively query, filter, and inspect node metadata without needing rigid query languages.

[46] Tomáš Pluskal, Santiago Castillo, Alejandro Villar-Briones, and Matej Orešič. MZmine 2: modular framework for processing, visualizing, and analyzing mass spectrometry-based molecular profile data. *BMC Bioinformatics*, 11:395, 2010.

Presents MZmine 2, a major architectural upgrade to the MZmine software suite. Featuring a highly extensible, modular Java framework, it provides advanced algorithms for mass spectrometry data peak picking, chromatogram building, retention time alignment, isotope grouping, and metabolite identification across complex metabolomics and proteomics datasets.

[47] Hasso L. Röst, Timo Sachsenberg, Stephan Aiche, Chris Bielow, Hendrik Weisser, Fabian Aicheler, Sandro Andreotti, Hans-Christian Ehrlich, Petra Gutenbrunner, Erhan Kenar, Xiao Liang, Sven Nahnsen, Lars Nilse, Julianus Pfeuffer, George Rosenberger, Marc Rurik, Uwe Schmitt, Johannes Veit, Mathias Walzer, Andreas Hildebrandt, Knut Reinert, and Oliver Kohlbacher. OpenMS: a flexible open-source software platform for mass spectrometry data analysis. *Nature Methods*, 13(9):741–748, 2016.

Describes OpenMS, a comprehensive and extensible open-source C++ software library and framework for mass spectrometry data management, analysis, and visualization. It provides a robust suite of algorithms and ready-made pipelines for proteomics, metabolomics, and targeted data processing (such as SWATH-MS), supporting standard file formats and facilitating reproducible computational workflows.

[48] Robin Schmid, Daniel Petras, Louis-Felix Nothias, Mingxun Wang, Allegra T. Aron, Annika Jagels, Hiroshi Tsugawa, et al. Ion identity molecular networking for mass spectrometry-based metabolomics in the GNPS environment. *Nature Communications*, 12:3832, 2021.

This article presents Ion Identity Molecular Networking, a computational framework that links related ion species within mass spectrometry datasets to improve metabolite annotation and clustering.

[49] Bastian Schulze, Youngjoon Jeon, Sarit Kaserzon, Amy L. Heffernan, Pradeep Dewapriya, Jake O'Brien, Maria Jose Gomez Ramos, Sara Ghorbani Gorji, Jochen F.

Mueller, Kevin V. Thomas, and Saer Samanipour. An assessment of quality assurance/quality control efforts in high resolution mass spectrometry non-target workflows for analysis of environmental samples. *TrAC Trends in Analytical Chemistry*, 133:116063, 2020.

This article provides a comprehensive overview of quality assurance and quality control measures in non-target analysis workflows, highlighting the importance of robust QA/QC practices to ensure reliable results in environmental sample analysis.

[50] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

This paper proposes a five-level confidence framework for identifying molecules in high-resolution mass spectrometry. The system is widely used in non-targeted analysis and provides standardized reporting for tentative and confirmed molecular identifications.

[51] Robert Sedgewick. *Algorithms in C, Parts 1–4: Fundamentals, Data Structures, Sorting, Searching*. Addison-Wesley Professional, Boston, MA, 3rd edition, 2002.

A definitive, highly accessible text covering foundational data structures and core algorithms implemented in C. It bridges theoretical analysis with practical implementation details, offering extensive empirical evaluations, concrete code examples, and hundreds of exercises for learning sorting, searching, and fundamental algorithmic techniques.

[52] Hiroshi Tsugawa, Tomas Cajka, Tobias Kind, Yan Ma, Brendan Higgins, Kazutaka Ikeda, Mitsuhiro Kanazawa, Jean VanderGheynst, and Masanori Arita. Ms-dial: Data-independent ms/ms deconvolution for comprehensive metabolome analysis. *Nature methods*, 12, 05 2015.

MS-DIAL is a software tool designed for data-independent acquisition (DIA) mass spectrometry, enabling comprehensive analysis of complex metabolomes by deconvoluting overlapping MS/MS spectra.

[53] Hiroshi Tsugawa, Kazutaka Ikeda, Mayumi Tanaka, Yukihiro Yamada, Nao Kawashima, Takaaki Mano, Tomomi Sugimoto, Sadayuki Kita, Masaru Tomita, Takaaki Nishioka, Yasumasa Arita, and Masanori Arita. A lipidome atlas in MS-DIAL 4. *Nature Biotechnology*, 38(10):1159–1163, 2020.

Introduces the MS-DIAL 4 software framework featuring a comprehensive lipidome atlas and in-silico tandem mass spectral libraries. It establishes universal data-processing tools for untargeted lipidomics, enabling robust annotation and structural characterization across diverse mass spectrometry platforms.

[54] Hiroshi Tsugawa, Rik Kessner, Atsushi Fukushima, Yukihiro Saito, Yoshihiro Suwa, Masanori Arita, Oliver Fiehn, and Hiroyuki Arai. Identifying metabolites by integrating metabolome databases with mass spectrometry cheminformatics. *Nature Methods*, 15(1):53–56, 2018.

Presents computational workflows and cheminformatics tools that integrate public metabolome databases with tandem mass spectrometry (MS/MS) data. The approach enhances compound identification confidence in untargeted metabolomics by combining spectral matching, in-silico fragmentation, and structural database searches.

[55] Hiroshi Tsugawa, Tomoya Nakabayashi, Tetsuya Mori, Yukihiro Yamada, Mayumi Tanaka, Yukihiro Sugimoto, Takeshi Bamba, Eiichiro Fukusaki, and Masanori Arita. A cheminformatics approach to characterize metabolomes in stable-isotope-labeled organisms. *Nature Methods*, 16(4):295–298, 2019.

Presents a cheminformatics-driven computational approach for untargeted plant metabolomics utilizing stable isotope labeling. The framework integrates mass spectrometry data processing with isotopic pattern recognition to improve formula assignment, metabolite annotation, and flux tracing in complex biological systems.

[56] Mingxun Wang, Jeremy J. Carver, Vanessa V. Phelan, Laura M. Sanchez, Neha Garg, et al. Sharing and community curation of mass spectrometry data with global natural products social molecular networking. *Nature Biotechnology*, 34(8):828–837, 2016.

This article discusses the use of social molecular networking for the sharing and curation of mass spectrometry data, highlighting its potential to enhance collaboration and data accessibility in the field of natural products research.

[57] Xiaoping Wang, Chuanfei Wang, Tingting Zhu, Ping Gong, Jianjie Fu, and Zhiyuan Cong. Persistent organic pollutants in the polar regions and the tibetan plateau: A review of current knowledge and future prospects. *Environmental Pollution*, 248:191–208, 2019.

This review highlights the need for continued monitoring and research on POPs in polar regions, particularly in the context of climate change and its impact on contaminant transport and fate.

[58] Xiaoping Wang, Tandong Yao, Zhiyuan Cong, Jiping Yan, Shichang Kang, and Lulun Ji. Distribution of persistent organic pollutants in soil and grasses around mt. qomolangma, china. *Archives of Environmental Contamination and Toxicology*, 52(2):153–162, 2007.

This study investigates the distribution of persistent organic pollutants (POPs) in soil and grasses around Mt. Qomolangma, highlighting the impact of altitude and environmental factors on contaminant accumulation.

[59] Frank Wania and Donald Mackay. Peer reviewed: Tracking the distribution of persistent organic pollutants. *Environmental Science & Technology*, 30(9):390A–396A, 8 1996. PMID: 21649427.

This article discusses the distribution and fate of persistent organic pollutants (POPs) in the environment, emphasizing the need for effective tracking and management strategies.

[60] Douglas B. West. *Introduction to Graph Theory*. Prentice Hall, Upper Saddle River, NJ, 2nd edition, 2001.

A widely adopted, classroom-tested textbook known for its clarity, pedagogical structure, and comprehensive coverage of both classical and modern graph theory. It provides rigorous proofs, historical context, and an extensive collection of exercises ranging from fundamental algorithms to advanced structural topics.

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